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Verification-first autonomous catalysis: large language models as infrastructure for mechanism, computation, and experiment

Yan Liu¹ & Pengfei Ou^{1,2}✉

Large language models are shifting catalysis from prediction-first pipelines to workflow-level coordination of hypotheses, computation, and experiments. In this Perspective, we argue that autonomous catalysis requires closing both representation and workflow gaps through a verification-first paradigm, in which large language model agents orchestrate heterogeneous tools, ground claims in auditable evidence, and use uncertainty-aware validation. We further propose system-level metrics to benchmark reliable autonomous catalysis.

Catalysis lies at the core of the modern chemical industry and sustainable energy technologies, enabling processes such as fuel production, carbon utilization, and environmental remediation^{1,2}. Despite decades of progress, catalyst discovery and optimization remain constrained by multiscale complexity: mechanisms span multiple length and time scales, active sites evolve under operando conditions, and performance emerges from subtle structure–environment coupling³. These features limit purely empirical trial-and-error and make isolated screening insufficient.

Recent advances in artificial intelligence (AI), particularly machine learning (ML), have accelerated catalysis by enabling high-throughput property prediction and data-driven screening^{4–7}. However, most approaches remain task-oriented and prediction-first, focusing on adsorption energies, barriers, or material properties within narrow representations and under fixed assumptions. Consequently, these models often struggle to compare mechanistic hypotheses, adapt to the relevant environmental changes, or support the iterative decisions central to catalyst development.

Large language models (LLMs) introduce different opportunities. As foundation models, they can integrate heterogeneous information, coordinate tools, and generate structured hypotheses and plans^{8,9}. Unlike conventional ML models, LLMs can integrate textual knowledge with computational outputs, experimental observations, and domain constraints within a single decision context^{10,11}. This capability positions LLMs as potential coordinators of catalytic research workflows¹².

In this Perspective, we argue that the most consequential role of LLMs in catalysis is to enable verification-first catalytic reasoning, in which language-generated hypotheses are treated as provisional claims that must pass predefined admissibility, consistency, and provenance checks before they can update the workflow state or trigger downstream decisions. The

representation gap arises because mechanistic correctness is defined at the level of pathway objects (reaction networks, transition states, and kinetics), which are rarely encoded explicitly in current pipelines. The workflow gap arises because discovery is procedural and iterative, requiring decisions about what to model or measure next under changing conditions. We outline design requirements for physics-grounded, experiment-compatible LLM agents emphasizing Verifier checks, evidence ledgers, and uncertainty gating and propose system-level evaluation criteria, including auditability/provenance, robustness under distribution shift, and closed-loop sample efficiency.

From catalyst prediction to catalytic reasoning

Conventional AI has improved catalytic property prediction and screening, but these advances do not automatically translate into mechanistic understanding or actionable decisions. In catalysis, “correctness” is often defined at the level of pathway competition, rate-/selectivity-controlling steps, and condition dependence (potential, solvation, coverage), rather than by single-point accuracy. This motivates a transition from prediction-first pipelines toward reasoning-first, workflow-integrated systems that can formulate, test, and revise mechanistic hypotheses through tool-grounded evidence.

Limitations of current AI paradigms in catalysis

ML models can predict adsorption energies, reaction barriers, and selectivity trends across large chemical spaces, enabling rapid screening and prioritization^{5,13,14}. Graph neural networks (GNNs), kernel regressors, and descriptor-based models achieve strong accuracy on benchmark datasets and are widely used for screening^{13,15,16}. Despite these successes, prevailing paradigms remain limited by a task-specific, prediction-first design

¹Department of Chemistry, National University of Singapore, Singapore, Singapore. ²NUS Artificial Intelligence Institute, National University of Singapore, Singapore, Singapore. ✉e-mail: pengf.ou@nus.edu.sg

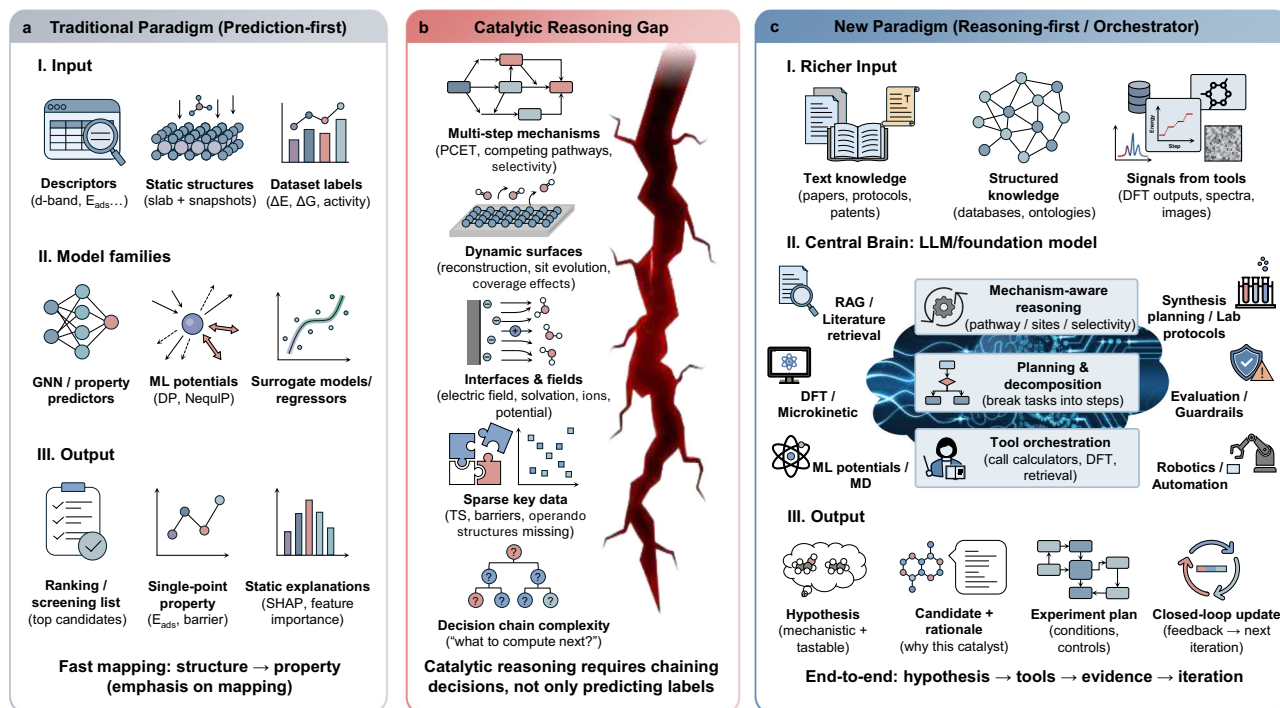


Fig. 1 | Paradigm shifts from prediction-first models to reasoning-first catalytic workflows. **a** Conventional AI pipelines that learn a largely static mapping from structures or descriptors to properties, primarily for screening and ranking. **b** The reasoning gap in catalysis arises from multi-step mechanisms, dynamic surfaces, and

interface-dependent operating conditions, together with sparse and inconsistent mechanistic labels. **c** Foundation models, including LLMs, act as coordinators that integrate heterogeneous knowledge and tool feedback to generate testable hypotheses, mechanistic rationales, and closed-loop plans.

(Fig. 1a)^{17,18}. Most models approximate a fixed mapping, such as structure-to-energy or composition-to-activity, under narrow assumptions. Transfer typically degrades when reaction conditions, active surfaces, or operative mechanisms change, limiting extrapolation beyond the training domain.

These limitations can be grouped into two coupled gaps that constitute the catalytic reasoning gap (Fig. 1b). First, there is a representation gap: catalysis is governed by competing pathways and transition states, yet many ML workflows output property estimates without explicitly representing mechanisms as falsifiable objects^{19–21}. Interpretability is often inferred indirectly rather than encoded as structured pathway hypotheses that can be validated, refuted, or updated^{22–24}. As a result, model outputs can remain descriptive, making it difficult to compare mechanistic alternatives, identify rate/selectivity control, or reason about condition-dependent shifts.

Second, there is a workflow gap: catalyst discovery is procedural and iterative, whereas ML models are typically passive modules^{19–21}. Real discovery cycles through hypothesis generation, modeling, testing, and revision; however, humans still orchestrate tool execution, data acquisition, constraint handling, and interpretation. This human-centered coordination limits scalability and makes end-to-end automation difficult to sustain. The mismatch is amplified by operando dynamics, where most models assume static representations^{25–27}. Site reconstruction, coverage fluctuations, and pathway competition can vary with potential, temperature, pressure, and the electrolyte environment, so training on equilibrium structures or isolated intermediates may miss key effects and bias mechanism identification and ranking.

Taken together, prediction-first paradigms are insufficient for mechanism-driven discovery or autonomy because they neither encode mechanisms as causal, testable objects nor provide a system-level procedure for deciding what to do next under changing conditions.

Catalysis as a stress test for LLM agents?

LLMs introduce a different paradigm for catalytic complexity²⁸. Rather than functioning only as task-specific predictors, LLMs can serve as reasoning controllers that integrate heterogeneous information and coordinate

multistep workflows²⁹. Recent work has also begun to explore agentic AI systems for chemical discovery and catalysis, highlighting the potential of language-driven agents to coordinate computational tools and experimental workflows. These studies emphasize the importance of integrating reasoning, tool use, and domain knowledge within scientific workflows rather than treating AI models purely as predictive components^{30,31}. This is particularly consequential in catalysis, where knowledge spans literature, simulations, experiments, and mechanistic hypotheses that must ultimately be tested against energetics and kinetics.

First, LLMs can support multi-source knowledge integration by placing textual knowledge (proposed mechanisms, empirical rules) alongside numerical outputs (energies, barriers, and kinetic parameters from density functional theory (DFT)) and experimental observations (operando spectroscopy, selectivity trends, and stability signatures) with a single decision context²⁸. If claims are anchored to attributable retrieval and tool-generated evidence, LLMs can help construct coherent mechanistic hypotheses across modalities rather than treating these inputs as isolated streams^{10,11}.

Second, LLM-based systems can enable procedural reasoning and planning through task decomposition and tool-calling: identifying plausible active sites, enumerating competing reaction pathways, selecting appropriate computational tools, and interpreting simulation outputs as evidence arrives^{32,33}. This mode of operation aligns with the iterative decision-making process used by catalysis experts and directly targets the workflow gap of conventional ML models.

Third, when embedded within agent-based architectures, LLMs can shift AI from passive prediction to active workflow coordination. An LLM-centered agent can decide when to run simulations, retrieve prior results, refine assumptions, or request additional data, while maintaining an explicit task state and rationale for each action^{11,34}. This transforms AI into an infrastructural layer that manages hypothesis tracking, tool orchestration, and experiment planning within a single loop (Fig. 1c).

However, catalysis also exposes fundamental constraints that LLM agents must address. Mechanism-aware reasoning must be verification-first: LLMs do not intrinsically enforce physical constraints, and therefore

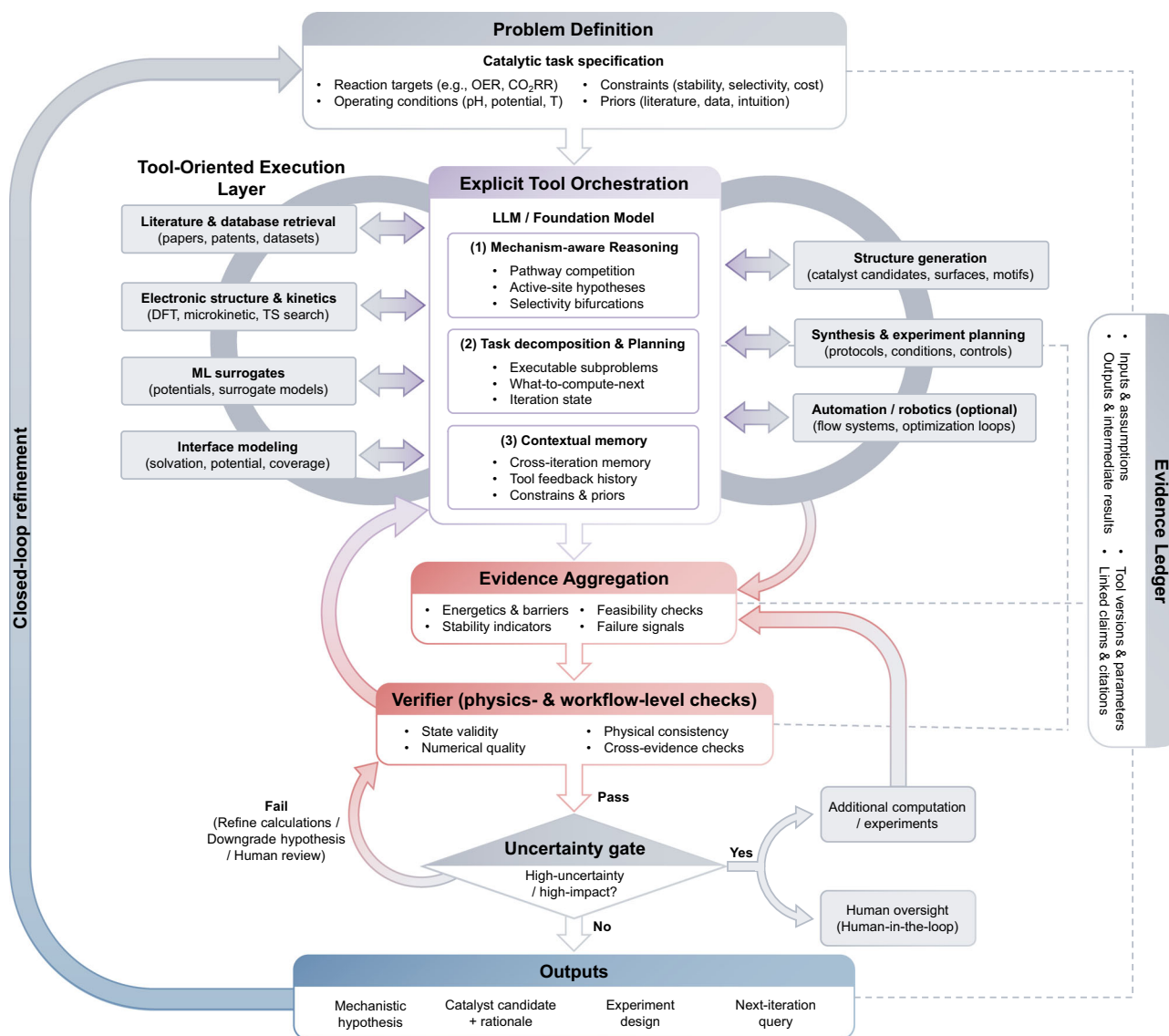


Fig. 2 | LLM-centered catalytic workflow with verification-first tool orchestration. A catalytic task is specified with reaction targets, operating conditions, and constraints. The LLM core performs mechanism-aware reasoning (pathway competition, active-site hypotheses, selectivity bifurcations), decomposes the task into executable subproblems, and maintains contextual memory across iterations. Specialized tools, including literature and database retrieval, electronic-structure and kinetic calculations (DFT, transition-state search, microkinetic modeling), interface modeling (solvation/potential/coverage), ML surrogates, structure generation, and experimental planning/automation—are invoked through explicit tool orchestration. A Verifier module implements a minimal verification policy that evaluates mechanistic claims using state-validity checks, numerical-quality checks, physical-consistency checks, and cross-evidence comparison, ensuring that

hypotheses are admitted into the workflow only when supported by tool-generated evidence. Representative checks include stoichiometry and charge conservation, thermodynamic consistency, barrier sanity, cross-tool agreement, and failure detection (e.g., non-converged calculations). When inconsistencies are detected, the system triggers structured responses such as refinement of calculations, hypothesis downgrading, or escalation to human oversight. All intermediate results and scientific claims are recorded in an evidence ledger that tracks provenance (inputs, tool versions, parameters, and outputs) to ensure auditability and reproducibility. An uncertainty gate routes high-uncertainty or high-impact decisions to additional computation or experiments, or to human oversight, enabling a closed-loop cycle that iteratively refines mechanistic hypotheses, catalyst candidates with rationale, experimental designs, and next-iteration queries.

language-level explanations may diverge from energetically or kinetically consistent mechanisms unless grounded in domain-specific tools. Therefore, LLM outputs should be treated as hypotheses that are constrained by physics-informed representations (reaction networks, step libraries, transition state (TS) data, interface descriptors) and validated by domain-specific tools³⁵.

LLM as the central coordinator in catalytic workflows

Building on the preceding discussion, LLM-centered systems can address the representation gap by elevating mechanisms to structured, testable objects, and address the workflow gap by enabling stateful, tool-mediated “reason-act-revise” loops with delegated decision authority over action

selection. Crucially, these systems must be verification-first: hypotheses proposed in language are treated as provisional claims and are admitted into the workflow state only when their supporting evidence satisfies predefined admissibility, numerical-quality, and physical-consistency criteria enforced by a verification layer (Fig. 2).

Mechanism-aware reasoning: pathways, active sites, and selectivity

The central challenge is not merely improving pointwise prediction accuracy but rather closing the representation gap: catalytic decisions depend on pathway competition, rate-/selectivity-controlling steps, and condition dependence, which must be represented as falsifiable mechanistic objects

rather than inferred post hoc from isolated property predictions. Physics-based modeling (e.g., DFT energetics/barriers and microkinetics) links elementary steps to macroscopic rates and selectivity, but practical progress relies on procedural choices what to model next, which interface assumptions to adopt, and how to reconcile computation with experiment³⁶.

In this setting, LLMs are best viewed as a coordination layer that maintains and updates an explicit hypothesis set: active-site motifs, competing pathways, and selectivity bifurcations, each associated with assumptions (facet, coverage, potential/solvation model) and evidence status. For branching search, tree/graph formalisms enable parallel comparison, constraint checking, and iterative refinement³⁷. Concretely, mechanism-aware representations can be organized as reaction graphs in which nodes encode catalytic states (site + adsorbate + environment descriptors), and edges encode elementary steps annotated with energetics/barriers, provenance, and validity checks, making “mechanistic correctness” measurable and revisable rather than narrative.

To enable comparison across hypotheses, catalytic mechanisms can be represented as structured mechanism objects rather than narrative descriptions. A minimal schema contains three components: species states, site descriptors, and elementary steps. Species states encode the catalytic configuration as a tuple consisting of surface motif, adsorbate identity, coverage, and environment descriptors (e.g., potential, solvation, electrolyte conditions). Site descriptors capture the identity and evolution of the active center, including local coordination environment, oxidation state, and reconstruction motifs when relevant. Elementary steps connect states through chemically meaningful transitions, annotated with energetics, transition states, and provenance from computational or experimental sources.

Dynamic active-site evolution can be represented through state transitions within mechanism-aware reaction graphs, where reconstruction, oxidation-state changes, or adsorbate-induced restructuring are encoded as state updates rather than external annotations. For example, surface reconstruction, oxidation-state changes, or adsorbate-induced restructuring are encoded as state updates rather than external annotations, allowing mechanisms with evolving active sites to be compared within the same representation. In practice, competing mechanisms are evaluated by combining thermodynamic feasibility, kinetic accessibility, and consistency with experimental observations. Under partial evidence, hypotheses can be ranked using progressively higher-fidelity evaluation layers, for example, surrogate screening, electronic-structure energetics, transition-state refinement, and microkinetic modeling, while pruning pathways that violate physical constraints or contradict validated evidence. This representation allows mechanism comparison to remain falsifiable and evidence-driven even when the active site itself evolves under reaction conditions.

To keep mechanistic reasoning traceable and correctable, systems should rely on updatable external memory and tool logs rather than parametric recall. Retrieval-augmented generation (RAG) combines attributable retrieval with generation, enabling evidence-grounded synthesis across literature and internal records with citation-level provenance³⁸. Grounding can be further strengthened by constraining reasoning with structured, physics-bearing representations. Knowledge-graph-augmented LLMs can bind pathway proposals to structured reaction objects (species, steps, constraints) with explicit evidence links and attribution³⁵.

Critically, mechanism-aware reasoning must be verification-first: language-generated pathway hypotheses are treated as provisional until validated (or falsified) by physics-based tools (DFT/TS search, microkinetics, stability filters) and explicit consistency checks. This design turns mechanistic reasoning into a controlled hypothesis–test–update cycle, enabling reliable comparison of competing pathways and condition-dependent selectivity under a shared, auditable workflow state (Fig. 2).

Tool-oriented catalytic agents

The workflow impact becomes concrete when LLMs address the workflow gap by operating in a tool-oriented, stateful “reason–act–revise” loop: decompose tasks, invoke tools, consolidate evidence, and re-plan based on

validation outcomes and failure signals. Representative frameworks such as ReAct and Toolformer formalize auditable tool use and specify when tool outputs should be incorporated into subsequent reasoning^{32,39}.

A catalysis toolchain encompasses retrieval, electronic-structure/kinetic tools (DFT, TS search, microkinetic modeling), interface modeling (solvation, potential, coverage), ML surrogates, structure generation, and experimental planning (optionally automated or robotic)⁴¹. ChemCrow illustrates that controller-plus-tools orchestration is practically realizable⁴⁰. However, reliability in catalysis requires that tool use be coupled to verification-first control rather than free-form iteration. Accordingly, tool invocation should be gated by (i) pre-checks on assumptions and inputs (e.g., surface model validity, charge/stoichiometry, feasible coverage/potential ranges), (ii) runtime monitoring with explicit failure handling (e.g., nonconvergence, unstable structures, inconsistent energies), and (iii) post-checks that enforce physical consistency (thermodynamic ordering, barrier sanity, cross-tool agreement) and trigger revision when constraints are violated.

To support auditability and reproducibility, systems should record not only final answers but also an evidence ledger: retrieved sources, tool calls, parameters, versions, outputs, and decision justifications linked to each hypothesis update. In closed-loop settings, an uncertainty gate is essential: high-uncertainty or high-impact decisions should trigger additional computation/experiments or human oversight rather than automated execution. Self-driving-lab reviews emphasize plan–execute–validate–revise loops and traceable evidence as prerequisites for robust autonomy⁴⁹.

Accordingly, outputs should be scientific deliverables rather than a single score, including testable mechanistic hypotheses, candidate catalysts with rationale and assumptions, experimental designs with controls, and next-iteration queries that target dominant uncertainties. These outputs update the task state and constraints for the next cycle, enabling sustained integration of mechanistic reasoning with tool-grounded evidence within a closed-loop workflow.

In practice, verification-first workflows do not attempt to evaluate all hypotheses exhaustively. Instead, verification proceeds through a multi-fidelity ladder in which inexpensive models are used to triage candidate mechanisms before higher-cost validation is performed. For example, descriptor-based heuristics or surrogate models may first eliminate clearly implausible hypotheses, followed by selective electronic-structure calculations to evaluate key intermediates. Only a small subset of surviving candidates may then undergo transition-state refinement or microkinetic modeling. Such staged verification prioritizes mechanism-disambiguating tests rather than exhaustive exploration of the hypothesis space. For example, in an OER system competing hypotheses such as AEM or lattice-oxygen mechanisms may first be screened using descriptor-level energetics before targeted transition-state calculations are performed for key steps.

To make verification-first control operational rather than rhetorical, each mechanistic claim should be evaluated under a minimal verification policy specifying (i) the claim type, (ii) admissible evidence types, (iii) required tool outputs, (iv) physical and numerical consistency checks, and (v) failure responses.

At minimum, four classes of checks are required. State-validity checks confirm that the catalytic state being evaluated is physically well defined, including stoichiometry, charge/spin balance, adsorbate placement, surface termination, and declared environment descriptors such as coverage, potential, and solvation model. Numerical-quality checks ensure that tool outputs are reliable for downstream reasoning, including convergence status, force thresholds, and reproducibility across equivalent calculations. Physical-consistency checks test whether the resulting energetics and pathways satisfy basic constraints such as mass and charge conservation, thermodynamic cycle consistency, and barrier sanity. Cross-evidence checks compare simulation outputs with literature reports, operando characterization, and kinetic trends to identify whether a claim is supported, contradicted, or underdetermined.

Failure in these checks does not necessarily invalidate a hypothesis but triggers a structured response: targeted refinement (for example, rerunning

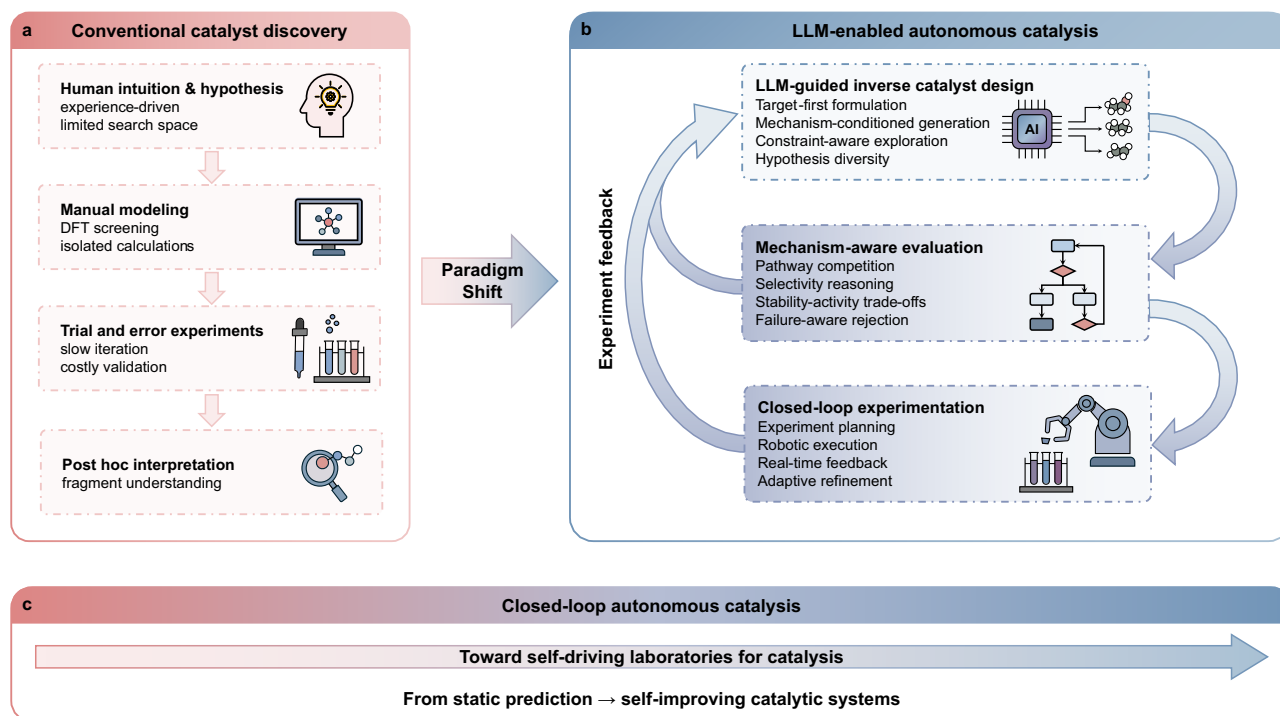


Fig. 3 | Emerging paradigms toward autonomous catalysis enabled by LLM-centered, verification-first systems. **a** Conventional catalyst discovery as a sequential, intuition-driven workflow with fragmented handoffs between modeling, experimentation, and interpretation. **b** LLM-enabled paradigms that support inverse catalyst design, mechanism-aware evaluation, and closed-loop

experimentation by connecting hypothesis generation, computational screening, and experimental feedback into an integrated workflow. **c** Summary of the overarching transition from static prediction to self-improving catalytic systems, emphasizing continuous refinement of hypotheses and decision-making through feedback between reasoning, tools, and experiments.

a calculation with improved interface models), claim downgrading to a lower-confidence hypothesis, or escalation to human review when the decision has high experimental or computational cost. For instance, a mechanistic claim should not be admitted if the underlying electronic-structure calculation remains unconverged, if the active site reconstructs into a different motif, or if the predicted rate-determining step contradicts the computed free-energy landscape.

Emerging paradigms toward autonomous catalysis

Autonomous catalysis emerges when two advances co-develop: (i) mechanism-aware representations that make hypotheses falsifiable (closing the representation gap), and (ii) verification-first, tool-orchestrated workflows that decide what to do next under uncertainty (closing the workflow gap). Building on the reference architecture in Fig. 2, we highlight two converging paradigms, LLM-guided inverse discovery and closed-loop experimentation, that convert language-level proposals into auditable, tool-grounded action.

LLM-guided catalyst design and inverse discovery

Conventional catalyst discovery largely follows a forward paradigm: propose candidate structures first, then evaluate activity, selectivity, and stability through computation and experiments, and iterative refine⁴⁰. Despite scaling relations, volcano analyses, and microkinetics, the practical logic often remains “structure-first, evaluate-later” (Fig. 3a)⁴¹, which can underutilize mechanistic knowledge and make it difficult to explore hypothesis diversity under constraints.

Recent generative work reframes discovery as target-first design: specify objectives and constraints upfront, then generate candidates for subsequent evaluation. In electrocatalysis, inverse-design frameworks combine surface generation, prediction, and optimization to map objectives to candidates (e.g., CO₂RR-relevant surfaces)^{42,43}. Related work on motif/environment generation further supports “inverse generation + interpretable constraints” at the atomic scale⁴⁴.

Within this landscape, LLMs can elevate inverse design from latent-space optimization to constraint-expressible, mechanism-actionable, diversity-aware hypothesis generation⁴⁵. Specifically, LLM agents can translate high-level goals (activity, selectivity, stability, cost, poison tolerance, synthesis feasibility) into structured constraints and mechanism-aware points, e.g., excluding pathways known to over-oxidize, enforcing stability windows, or favoring motifs that enable a desired selectivity bifurcation. This shifts generative design toward hypothesis diversity with explicit rationale, rather than single-candidate optimization. Recent reviews in catalysis and materials increasingly position LLMs, together with automation and ML, as enabling components for goal-driven discovery⁴⁶.

Crucially, this paradigm does not require LLMs to “replace physics.” In a verification-first workflow, LLM outputs are treated as provisional hypotheses that must be filtered or falsified by physics-based evaluation (DFT energetics/barriers and microkinetic pathway competition), so that “mechanism-conditioned generation” yields candidates that are not only plausible but testable (Fig. 3b). In practice, this implies (i) generating multiple mechanistic hypotheses per candidate (active-site motifs, competing pathways, selectivity control points), (ii) routing evaluation to the appropriate tools (including interface-aware modeling when needed), and (iii) updating generation constraints based on validated failure modes (e.g., instability, unfavorable barriers, incorrect selectivity regime). This closes the loop between generation and mechanistic verification and directly addresses the representation gap by treating mechanisms as first-class objects within the design cycle.

Closed-loop catalysis: from hypothesis to experiment

Autonomous catalysis ultimately requires closed-loop execution: hypothesis formulation → candidate selection or generation → validation planning → experimental and/or computational execution → evidence aggregation with failure signals → adaptive refinement. Self-driving-lab analyses emphasize that coupling automation, orchestration software, and algorithmic decision-making enables sustained closed-loop iteration¹⁹.

Characterization as a mechanistic verification layer. In catalytic research, mechanistic hypotheses cannot be validated by computation alone and often require experimental characterization to confirm the identity and evolution of active sites under reaction conditions. Operando and in situ techniques, including spectroscopy, microscopy, and electrochemical probes play a crucial role in distinguishing between competing mechanisms and verifying whether predicted active states exist under reaction conditions. Within a verification-first workflow, characterization should therefore function as a complementary evidence source alongside computational tools. For example, operando spectroscopic signatures, catalyst reconstruction observed by microscopy, or isotope-labeling experiments can either support or falsify mechanistic hypotheses proposed by simulations or literature. In this sense, characterization data serve as high-value evidence entries in the evidence ledger and can trigger hypothesis revision or pruning when discrepancies arise. Integrating characterization into the closed-loop cycle thus strengthens the verification-first paradigm by linking mechanistic reasoning, computational validation, and experimentally observable phenomena.

Compared with earlier automation, a key differentiator of LLM-enabled loops is semantic continuity under verification-first control: the system maintains persistent representations of goals, assumptions, constraints, and uncertainty, and updates them as evidence arrives. As a proof of concept, Coscientist demonstrates that LLM-driven systems can execute end-to-end chains spanning planning, tool use, experiment execution, and interpretation, supporting the feasibility of maintaining language-level goals across a closed loop¹⁰. At the orchestration layer, ChemOS formalizes laboratories as coordinated systems with unified interfaces, enabling stateful scheduling and closed-loop control^{47,48}.

However, catalysis makes closed-loop autonomy uniquely demanding. The optimization target is typically multi-objective (activity, selectivity distribution, stability, cost) and strongly condition-dependent; moreover, mechanistic labels are sparse, and “ground truth” may shift as active sites reconstruct. Thus, robust loops must be designed around (i) uncertainty-aware decision policies, (ii) explicit constraint checking and failure handling, and (iii) auditable evidence trails that connect each action to tool outputs and provenance.

Closed-loop catalysis also hinges on information-gain-driven experiment selection (e.g., Bayesian optimization, active learning, and multi-objective optimization) rather than brute-force exploration⁴⁹. Automated platforms have demonstrated closed-loop self-optimization and scale-up in flow systems, directly instantiating plan → execute → feedback → refine cycles. For catalytic materials discovery, active learning has accelerated the discovery and optimization of electrocatalysis-relevant systems, consistent with the principle of maximizing insight per experiment^{50,51}. Multi-objective closed-loop optimization of catalyst formulations can converge within a few generations⁵². In verification-first settings, humans act as goal-setters and safety/audit gatekeepers rather than operators, and DMTA workflows highlight auditable decision chains and human-in-the-loop governance under cost and safety constraints⁵³.

Together, LLM-guided inverse design and closed-loop experimentation move catalyst discovery from sequential trial-and-error toward self-improving, mechanism-aware systems driven by iterative evidence and adaptive control (Fig. 3c). Importantly, autonomy should be defined at the system level the ability to propose hypotheses, select actions, validate with tools, and revise under uncertainty rather than by any single model’s predictive accuracy.

Key challenges unique to catalysis-oriented LLM systems

Scope and boundary conditions. Verification-first workflows are particularly valuable in catalytic systems characterized by substantial mechanistic ambiguity, strong operando dynamics, or frequent distribution shifts between computational models and experimental conditions. Examples include electrocatalysis and heterogeneous catalysis under reactive environments, where surface reconstruction, coverage effects, electrolyte interactions, or field effects can significantly alter the identity of active sites. In

such regimes, prediction-first screening may generate candidate lists without resolving the underlying reaction physics.

Conversely, in more stable catalytic systems governed by well-established descriptors and relatively fixed active sites, traditional prediction-first screening may remain effective. In these settings, surrogate models or descriptor-based rankings can already provide reliable guidance for materials selection. Verification-first should therefore be viewed not as a replacement for predictive screening, but as a complementary paradigm that becomes particularly powerful when mechanistic uncertainty limits the transferability of purely predictive models.

Despite growing interest in LLMs as central coordinators in catalytic research workflows, their deployment in catalysis faces challenges that are sharper than in many other scientific domains. Because catalysis is multi-scale, mechanism-driven, and experimentally constrained, LLM systems must be physically grounded, controllable, and experiment-compatible^{54–56}. Importantly, these challenges are not solved by model scale alone: they require closing both the representation gap (mechanisms as falsifiable objects) and the workflow gap (tool-mediated decision procedures), implemented through verification-first controls (Verifier, evidence ledger, and uncertainty gate).

Mechanistic grounding and physical fidelity

A primary challenge is insufficient mechanistic grounding (Fig. 4a). Catalytic performance depends on competing elementary steps and the TS landscape that governs the rate and selectivity⁵⁷. Without explicit pathways and TS information, outputs remain descriptive rather than falsifiable. Even in well-studied systems, microkinetic conclusions can change under common modeling assumptions, reinforcing that correlation does not imply causation. This gap becomes sharper at electrified interfaces, where operative physics depends on charge redistribution, interfacial electric fields, and solvent- and ion-mediated stabilization. If charge, field, and potential are weakly represented, the system may systematically misrank barriers or misattribute selectivity because solvation and interfacial structure can reshape both energetics and kinetics. Comprehensive interface-focused reviews emphasize that electrocatalytic modeling must account for environment-dependent effects (electrostatic field, solvation, ions, and potential) to remain physically meaningful⁵⁸.

A closely related catalysis-specific challenge is the capture of dynamic and operando effects (Fig. 4b)²⁵. Many workflows still rely on static inputs, whereas operando catalysis often demands time-resolved evidence. Without time-resolved constraints, reasoning can drift toward static-site assumptions, missing reconstruction, and active-site evolution under working conditions. Operando/in situ studies emphasize that identifying the true active species under reconstruction is central to interpreting activity and stability.

Implication for verification-first design. To address this, LLM systems must couple language-level hypotheses to physics-bearing mechanism objects (reaction networks, step libraries, TS candidates, microkinetic models, and interface descriptors) and require tool-grounded validation before admitting mechanistic claims into the workflow state. For example, calculations that fail numerical convergence, produce unstable structures, or violate thermodynamic ordering should automatically trigger recalculation, model refinement, or hypothesis downgrading rather than being converted into mechanistic conclusions.

Generalization across catalysts, interfaces, and conditions

A second difficulty is limited cross-system generalization (Fig. 4c). Catalysis spans diverse catalyst classes and environments, where domain-specific overfitting is common and intuitive analogies can break down across materials families (e.g., MOFs, oxides, and alloys)^{59,60}. Moreover, failures often arise from interfacial and solvent mismatches, in which the same nominal chemistry behaves differently under distinct interfacial structures or electrolyte conditions. In data-driven settings, such gaps manifest as sensitivity to distribution shift. Large-scale out-of-distribution studies in

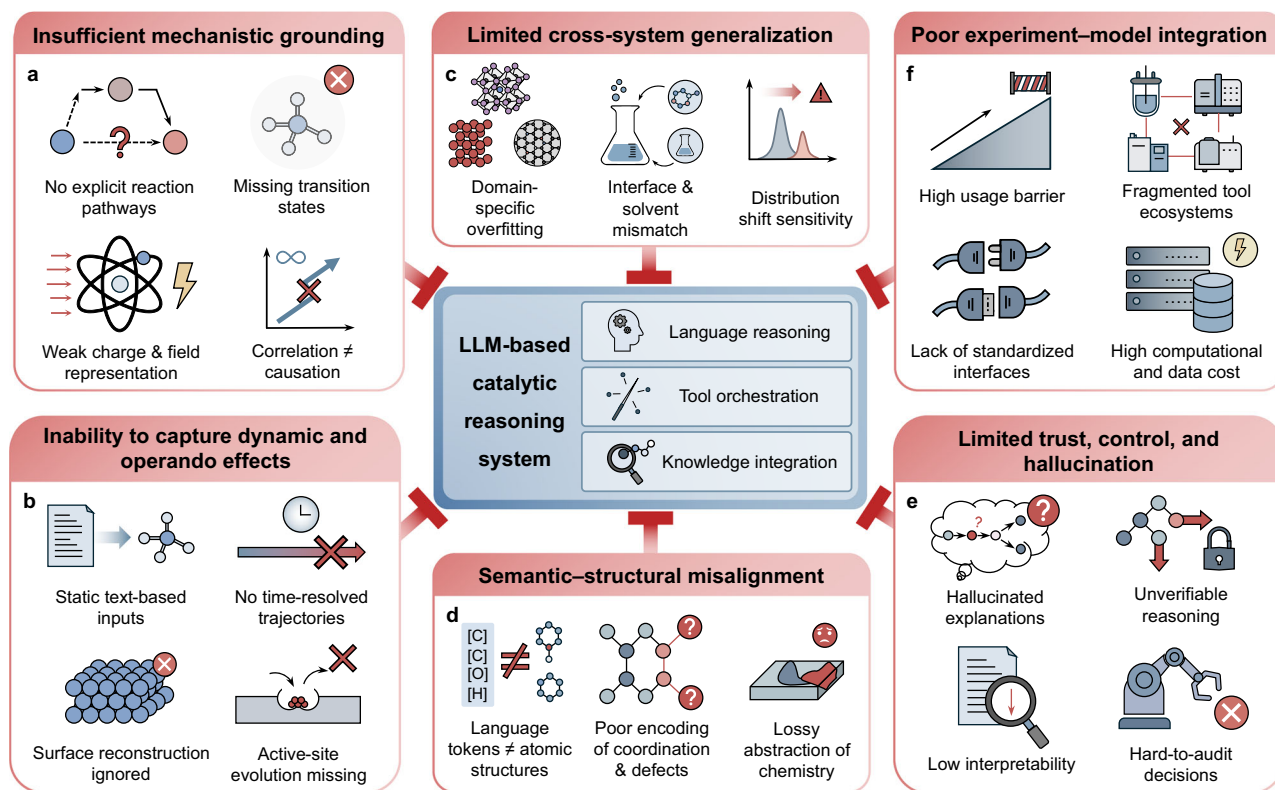


Fig. 4 | Key bottlenecks limiting LLM-driven autonomous catalysis. **a** Insufficient mechanistic grounding, where reaction pathways and transition states are not explicitly represented, and charge, field, and solvation effects are weakly captured, leading to correlation-driven conclusions. **b** Dynamic and operando complexity gap, where static inputs omit time-resolved trajectories, surface reconstruction, and active-site evolution under working conditions. **c** Limited cross-system generalization, including sensitivity to distribution shift and poor transfer across catalyst classes and environments such as alloys, oxides, MOFs, and interfaces. **d** Semantic

and structural misalignment, where language tokens do not faithfully encode coordination, defects, and interface structure, resulting in lossy abstraction and weak structure-function alignment. **e** Limited trust, control, and interpretability, including hallucinated explanations, unverifiable reasoning paths, low auditability, and hard-to-correct errors in iterative workflows. **f** Poor experiment-model integration and scalability, including high usage barriers, fragmented tool ecosystems, a lack of standardized interfaces, and high computational and data costs for physically realistic modeling.

materials ML show that conventional evaluations can overestimate real extrapolation, and performance often degrades when the test domain moves outside the training manifold⁶¹.

Equally important is semantic-structural misalignment (Fig. 4d). LLMs generalize primarily through patterns in semantic context, yet catalytic function is frequently dictated by precise local structure, coordination environments, defects, facets, and evolving ensembles⁶². Geometry-aware and structure-tokenization approaches underscore that semantics must align with 3D structure when structure is causal^{63,64}.

Implication for verification-first design. Robust autonomy, therefore, requires unified representations that bind text to structure/energetics/kinetics and evaluations that explicitly test transfer under distribution shifts. In practice, an uncertainty gate should detect when the current query departs from prior evidence (e.g., a new catalyst family, electrolyte regime, or interface model) and route decisions to additional computation/experiments rather than making confident extrapolations. Evidence ledgers should record the domain of validity (assumptions, environment descriptors, and tool settings) so that generalization claims are auditable and failures are attributable to specific mismatches.

Trust, control, and experimental compatibility

Beyond accuracy, adoption in experimental catalysis hinges on governance properties, including trust, control, and interpretability (Fig. 4e). LLMs can generate plausible explanations that lack sufficient evidential support, and their decision paths can be difficult to audit, leading to errors that are hard to correct, especially in closed-loop settings where automation can amplify

mistakes. Systematic surveys of hallucination and mitigation strategies document these failure modes and highlight the need for verification-centered workflows.

Finally, even a well-governed model can fail to translate into practice due to poor experimental integration and scalability (Fig. 4f). In many laboratories, LLM tooling faces significant adoption barriers and a fragmented tool ecosystem. The lack of standardized interfaces prevents robust handoff from language outputs to instrument control, execution, and data capture^{48,65}. Orchestration architectures such as ChemOS 2.0 target this systems-layer bottleneck by coordinating communication, data exchange, and instruction management among modular components^{47,48}. Standardization efforts, such as SiLA 2, similarly aim to improve interoperability through shared concepts and an open communication layer⁶⁶.

Scalability is further constrained by cost: realistic fidelity may require explicit interface sampling, coverage effects, uncertainty quantification, and microkinetics, exceeding routine closed-loop throughput. Therefore, practical systems need cost-aware tool scheduling, uncertainty-aware prioritization, and standardized experiment-model interfaces and data schemas.

Implication for verification-first design. Trust and experimental compatibility are system properties. A Verifier and evidence ledger operationalize “no claim without evidence,” while the uncertainty gate provides a principled trigger for deferral, additional validation, or human oversight. Together with standardized lab interfaces, these controls transform LLM outputs from persuasive text into auditable, executable research actions.

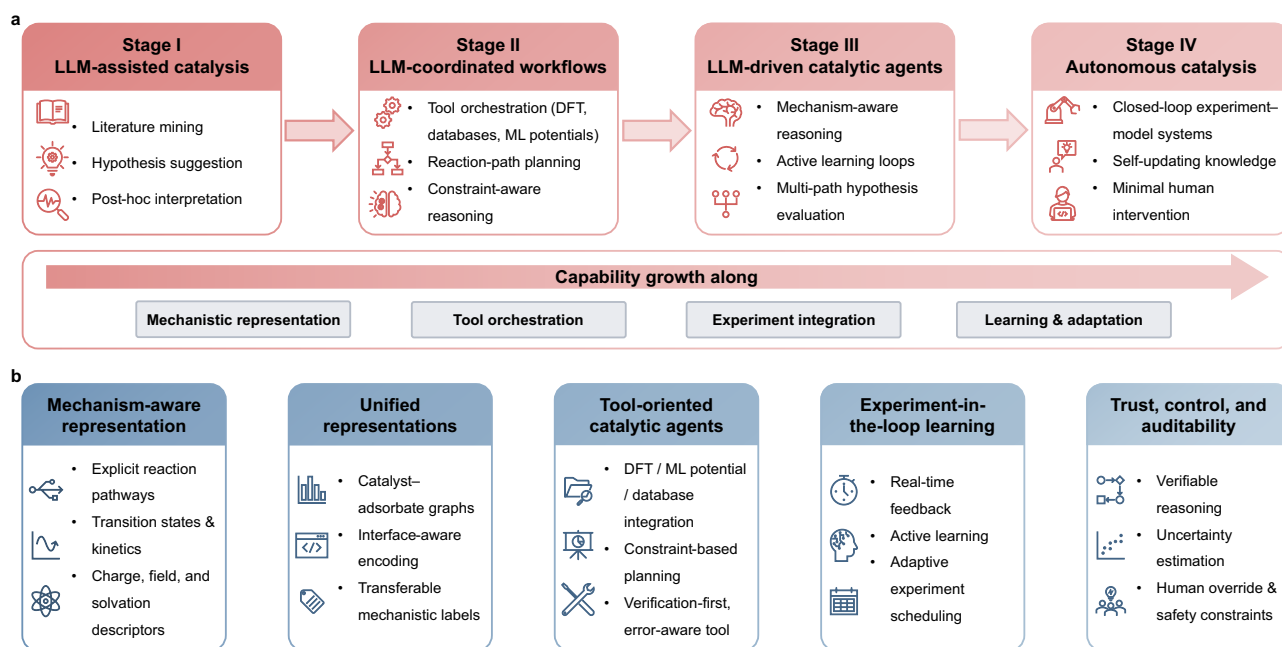


Fig. 5 | Roadmap toward autonomous catalysis enabled by LLM-centered, verification-first systems. a Evolution path from LLM-assisted catalysis to LLM-coordinated workflows, LLM-driven catalytic agents, and autonomous catalysis, highlighting capability growth along mechanistic representation, tool orchestration, experiment integration, and learning/adaptation. **b** Enabling pillars for this transition, including mechanistic grounding through mechanism-aware representation,

unified representations that bridge structural and semantic information, tool-oriented agents that integrate domain tools with constraint-aware planning, experiment-in-the-loop learning with real-time feedback and adaptive scheduling, and trust-aware system design with verifiable reasoning, uncertainty estimation, and human override and safety constraints.

Claim typing and evidence admissibility. To prevent “evidence laundering,” in which weak evidence is rhetorically amplified to support mechanistic claims, the evidence ledger must enforce explicit provenance and admissibility rules. In practice, claims should be typed according to their epistemic status (e.g., qualitative hypothesis, mechanistic proposal, rate-determining step assignment, or quantitative prediction), and each claim type should specify permitted evidence sources. For instance, literature reports or empirical heuristics may support exploratory hypotheses but are insufficient to validate pathway energetics or rate control without tool-grounded evidence such as electronic-structure calculations, kinetic modeling, or experimental characterization.

All evidence entries should include provenance metadata (source, tool version, input parameters, and calculation or experimental conditions) to ensure traceability. When conflicting evidence arises, for example, when literature suggests a pathway that contradicts computed energetics or kinetic trends the claim should be downgraded to an unresolved hypothesis rather than elevated to a conclusion. In such cases, the workflow should trigger targeted verification steps, such as additional calculations, sensitivity analysis, or experimental tests, until the discrepancy is resolved or the hypothesis is pruned. These admissibility rules ensure that the evidence ledger functions not merely as a record of results but as a governance layer controlling how evidence can justify scientific claims.

Epistemic grounding and structured representations

Recent discussions in AI-driven chemistry highlight an important explanatory trade-off: as AI systems increasingly mediate hypothesis generation and discovery, the epistemic burden shifts toward explanation, provenance, and responsibility⁶⁷. In this setting, verification-first orchestration can be viewed as an epistemic control mechanism that ensures scientific claims remain grounded in traceable evidence rather than persuasive inductive artifacts. The verifier → evidence ledger → uncertainty gate triad proposed in this work provides a concrete operationalization of this principle by enforcing admissibility rules for mechanistic claims, recording provenance across tool outputs, and

routing high-uncertainty decisions to additional validation or human oversight.

At the same time, the representation layer of autonomous catalysis must integrate heterogeneous sources of knowledge, including literature, simulations, experimental observations, and mechanistic constraints⁶⁸. Recent advances in heterogeneous knowledge graph representation learning suggest that structured relational representations can improve how heterogeneous signals are integrated and reasoned over. In the context of catalytic workflows, such approaches are complementary to the mechanism-object framework proposed here: graph-based learning can help discover statistical associations or latent structural motifs, whereas the mechanism-object layer enforces physically grounded constraints such as reaction topology, stoichiometry, transition-state connectivity, and microkinetic consistency. In this sense, autonomous catalysis may benefit from a hybrid paradigm in which machine learning captures patterns across heterogeneous data while physics-based mechanism representations define the admissible hypothesis space.

Outlook: What will truly enable autonomous catalysis

Autonomous catalysis will not be achieved by scale alone. A more plausible trajectory is staged: LLM-assisted → LLM-coordinated → LLM-driven agents → autonomous catalysis, where progress is evaluated by co-development of mechanistic representations (closing the representation gap), tool-orchestrated procedures (closing the workflow gap), and verification-first controls (Verifier, evidence ledger, and uncertainty gate) that make decisions auditable and safe (Fig. 5a). Reviews of self-driving laboratories emphasize that autonomy is operationally defined by the ability to plan, execute, and assimilate feedback within an integrated workflow, rather than by model size alone^{19,69}.

Stage I: LLM-assisted catalysis (productivity without execution authority)

In Stage I, LLMs are primarily used for literature mining, hypothesis suggestion, and post hoc interpretation, improving productivity but not

yet constituting an execution layer. The minimum viable capabilities are:

- Attributable retrieval and summarization with citation-level provenance (e.g., RAG over literature and lab records).
- Structured notetaking and memory that records assumptions, conditions, and key evidence rather than relying on parametric recall.
- Conservative interaction policies that avoid tool-free mechanistic claims when evidence is missing.

Stage II: LLM-coordinated workflows (tool use with verification gates)

Stage II emphasizes LLM-coordinated workflows in which an agent can call domain tools (DFT, databases, ML surrogates, microkinetics) via task decomposition and constraint-aware planning, while conclusions remain grounded in tool-generated evidence. The minimum viable capabilities are:

- Tool routing and job management (submit, monitor, recover from failures) across retrieval, computation, and analysis.
- Verifier-driven consistency checks (sanity checks on energetics/barriers, thermodynamic consistency, stability filters, nonconvergence handling).
- Evidence ledger logging for every claim and decision (inputs, tool versions, parameters, outputs, and links to retrieved sources).

Stage III: LLM-driven catalytic agents (multi-hypothesis reasoning under uncertainty)

Stage III requires a behavioral shift from “assisting” to “driving” iterative discovery. Agents must support mechanism-aware reasoning, active learning, and multi-hypothesis evaluation so alternatives can be proposed, tested, and updated under uncertainty. The minimum viable capabilities are:

- Mechanisms as first-class objects (reaction graphs/step libraries/TS candidates/microkinetic models) enabling hypothesis versioning and comparison.
- Uncertainty-aware decision policies that trigger additional validation (computation/experiments) or deferral when extrapolating across catalysts/interfaces/conditions.
- Information-gain-driven scheduling (active learning/multiobjective optimization/Bayesian optimization) tightly coupled to mechanistic uncertainty.

Stage IV: Autonomous catalysis (sustained closed-loop discovery with governance)

Stage IV, autonomous catalysis, is best defined as sustained closed-loop experiment–model systems with self-updating knowledge and minimal human intervention^{19,69}. Here, autonomy is a system property: the ability to propose hypotheses, select actions, validate with tools, and revise under uncertainty while remaining auditable and safe. The minimum viable capabilities are:

- Closed-loop integration of planning → execution → validation → revision with standardized experiment–model interfaces and data schemas.
- Governance and override mechanisms (safety constraints, budget constraints, human-in-the-loop triggers for high-impact decisions).
- Continuous learning with provenance (updating priors, step libraries, and surrogate models while preserving reproducibility).

Across stages, progress should be measured not only by catalytic outcomes, but by system-level metrics: (i) auditability and provenance (can every decision be traced to evidence?), (ii) robustness under distribution shift (how does performance degrade when catalysts/interfaces/conditions change?), and (iii) closed-loop sample efficiency (improvement per unit cost in experiments, computation, and human time).

Enabling pillars for verification-first autonomous catalysis

Several system capabilities jointly enable this evolution (Fig. 5b). First, mechanistic grounding depends on mechanism-aware representations that expose explicit reaction pathways, TS, kinetics, together with environment descriptors capturing charge, field, solvation, and potential^{70,71}. Without these causal objects, systems are vulnerable to correlation-driven reasoning, especially at electrified interfaces.

Second, robust transfer across catalysts, interfaces, and conditions requires unified representations that genuinely bridge language and structure, such as catalyst–adsorbate graphs, interface-aware encoding, and transferable mechanistic labels⁷².

Third, agents must be verification-first by design. Rather than trusting fluent text, systems should earn credibility through domain tools (DFT, ML surrogates, databases, microkinetics), explicit failure handling, and cross-tool consistency checks implemented via a Verifier and evidence ledger that operationalize “no claim without evidence”^{11,19}.

Fourth, experiment-in-the-loop learning requires adaptive scheduling that prioritizes expected information gain and reduces uncertainty where it matters for mechanism and decision-making^{73,74}. This capability is most effective when uncertainty estimates, and mechanistic sensitivity analyses directly inform the selection of which experiments or simulations to conduct next.

Finally, real laboratory deployment depends on trust, control, and auditability as governance properties: verifiable reasoning traces, uncertainty gating for high-impact decisions, standardized data schemas, and human override and safety constraints. Verification-centered protocols, such as chain-of-verification, provide concrete mechanisms to reduce unsupported claims and improve the auditability of LLM outputs.

Conclusions

LLMs are reshaping catalytic research by extending AI beyond prediction and screening. Rather than functioning as task-specific regressors, LLM-centered systems can integrate heterogeneous knowledge, coordinate tool use, and support multi-step reasoning across literature, computation, and experimental records. Accordingly, we argue that LLMs will matter most in catalysis as workflow infrastructure that earns credibility by linking hypotheses to verification-gated actions, rather than as a replacement for physics-based modeling or experiments.

At the same time, autonomous catalysis remains constrained by intrinsic catalytic complexity. Mechanistic grounding is essential: outcomes depend on pathways and transition states, as well as environment-dependent effects, and cannot be inferred from fluent text alone. Generalization is also nontrivial: apparent out-of-distribution success can reflect interpolation, and performance often degrades under genuine distribution shift across catalyst classes, interfaces, and operating regimes. Practical autonomy, therefore, requires verification-first closed-loop operation in which planning, execution, validation, and revision are integrated as a single capability, with tool evidence not narrative plausibility, determining which hypotheses survive.

Looking forward, progress will depend less on scale than on system design and integration. We propose three design principles for reliable autonomous catalysis:

(1) Mechanism as an object: represent hypotheses as structured, falsifiable mechanism objects (reaction graphs, TS candidates, microkinetic models) with explicit assumptions and versioning.

(2) Verification-first orchestration: embed domain tools within a controlled loop (Verifier + evidence ledger + uncertainty gate) that enforces physical consistency, handles failures, and attaches every claim to traceable evidence.

(3) Cost- and experiment-aware autonomy: couple decision policies to information gain under budgets and deploy standardized experiment–model interfaces that enable robust closed-loop execution.

Consistent with these principles, we advocate evaluating progress using system-level metrics not outcomes alone including auditability and provenance, robustness under distribution shift, and closed-loop sample

efficiency. By treating LLMs as infrastructure with verifiable interfaces to physics and experiments, catalytic discovery can shift from trial-driven iteration to adaptive, hypothesis-driven cycles that deliver actionable structure–function insights across catalysts and conditions.

Data availability

No datasets were generated or analysed during the current study.

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Author contributions

Y.L. and P.O. conceived the perspective. Y.L. conceived the overall framework and central arguments of the perspective, integrated perspectives from catalysis and large language models, and drafted the manuscript. Y.L. also designed the conceptual figures, including Figures 1 to 5, and coordinated revisions to ensure coherence between the text and graphical narratives. P.O. supervised the project, contributed to the conceptual development of the autonomous catalysis framework, provided critical feedback on structure and positioning, and substantially revised the manuscript for intellectual content. All authors discussed the key themes and interpretations presented in the review and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Pengfei Ou.

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